Function $f(x)$	Indefinite integral $\int f(x)dx$
$\frac{1}{\sqrt{a^2+x^2}}$, where a is a constant	$\log_e\left\|x+\sqrt{a^2+x^2}\right\|+C$
$\frac{1}{\sqrt{a^2-x^2}}$, where $a\neq 0$ is a constant	$\sin^{-1}\left(\frac{x}{a}\right)+C$
$\frac{1}{\sqrt{x^2-a^2}}$, where a is a constant	$\log_e\left\|x+\sqrt{x^2-a^2}\right\|+C$
$\sqrt{a^2+x^2}$, where a is a constant	$\frac{x\sqrt{a^2+x^2}}{2}+\frac{a^2}{2}\log_e\left\|x+\sqrt{a^2+x^2}\right\|+C$
$\sqrt{a^2-x^2}$, where a is a constant	$\frac{x\sqrt{a^2-x^2}}{2}+\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)+C$
$\sqrt{x^2-a^2}$, where a is a constant	$\frac{x\sqrt{x^2-a^2}}{2}-\frac{a^2}{2}\log_e\left\|x+\sqrt{x^2-a^2}\right\|+C$

9.2 Definite Integral as the Limit of a Sum

9.2.1 Riemann Integral

Consider a real-valued, bounded function $f(x)$ defined on the closed and bounded interval $[a,b]$, $a<b$. The function $f(x)$ need not have the same sign on $[a,b]$; that is, $f(x)$ may have positive as well as negative values on $[a,b]$. See Fig 9.2. Partition the interval $[a,b]$ into n subintervals $[x_0,x_1],[x_1,x_2],\cdots,[x_{n-2},x_{n-1}],[x_{n-1},x_n]$ such that

$$a=x_0<x_1<x_2<\cdots<x_{n-1}<x_n=b.$$

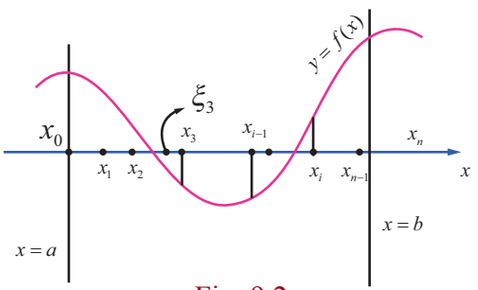


Fig. 9.2

In each subinterval $[x_{i-1},x_i], i=1,2,\cdots,n,$ choose a real number ξ_i arbitrarily such that $x_{i-1}\le\xi_i\le x_i$.

Consider the sum $\sum_{i=1}^{n} f(\xi_i)(x_i-x_{i-1}) = f(\xi_1)(x_1-x_0)+f(\xi_2)(x_2-x_1)+\cdots+f(\xi_n)(x_n-x_{n-1})$ ....(1)

The sum in (1) is called a **Riemann sum** of $f(x)$ corresponding to the partition $[x_0,x_1],[x_1,x_2],\cdots,[x_{n-1},x_n]$ of $[a,b]$. Since there are infinitely many values ξ_i satisfying the condition $x_{i-1}\le\xi_i\le x_i$, there are infinitely many Riemann sums of $f(x)$ corresponding to the same partition $[x_0,x_1],[x_1,x_2],\cdots,[x_{n-1},x_n]$ of $[a,b]$. If, under the limiting process $n\to\infty$ and $\max\left(x_i-x_{i-1}\right)\to 0$, the sum in (1) tends to a finite value, say A, then the value A is called the **definite integral** of $f(x)$ with respect to x on $[a,b]$. It is also called the **Riemann integral** of $f(x)$ on $[a,b]$ and is denoted by $\int_a^b f(x)dx$ and is read as the integral of $f(x)$ with respect to x from a to b. If $a=b$, then we have $\int_a^a f(x)dx=0$.

Chapter 9 Applications of Integration.indd 92 17-02-2020 17:03:18